Statistical Learning and Machine Learning

Exam Topics - January 2022

TOPIC 1

Consider that you obtained N=400 measurements of a data set $x_i \in \mathbb{R}^2$, plotted in Fig.1. The objective here is to model the observed data using a mixture of Gaussians model with $K = 2$ components. With this background, answer the following questions:

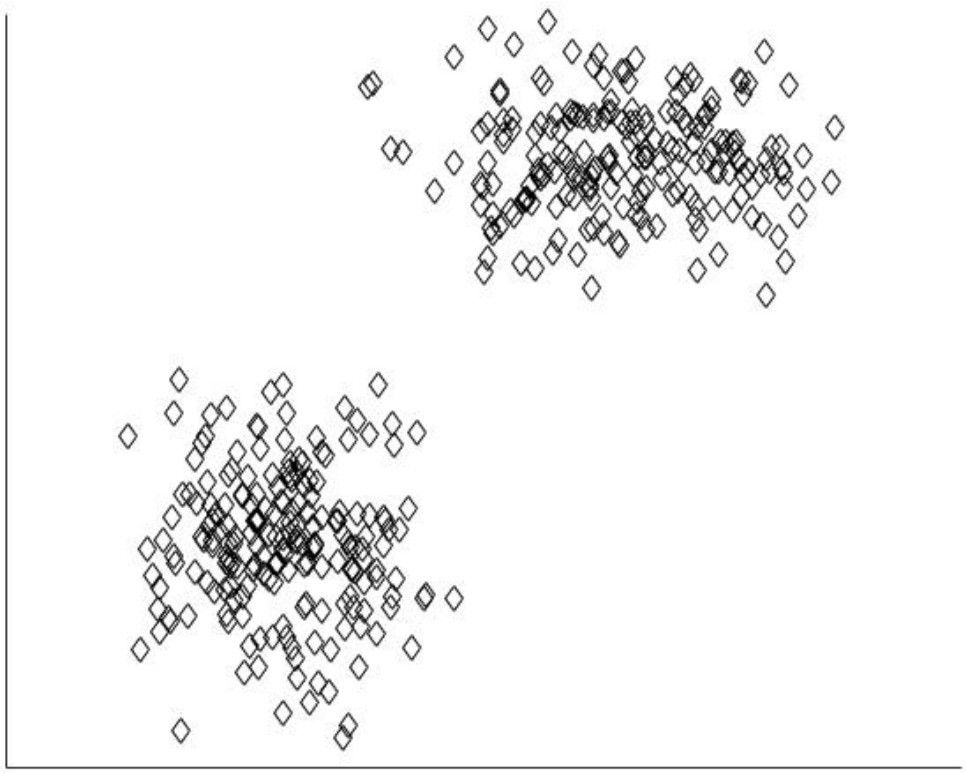

Fig. 1. The data points $x_i \in \mathbb{R}^2$ are plotted in 2D space as diamonds. It is desired to choose an appropriate distribution model to represent this data set as well as to obtain the parameters of the chosen model.

1) (Score: 20 %) List the parameters of the Gaussian mixture model that need to be estimated (from observed data) to fully specify the model.
2) (Score: 30 %) Assuming that the observed data points are i.i.d (independent and identically distributed) and that maximum likelihood estimation (MLE) is to be used for the estimation of model parameters, write the likelihood function that will be maximized within MLE?
3) (Score: 30 %) Briefly describe the methodology that you would adopt to estimate the mean vector $\boldsymbol{\mu}_k$ of the $k-$th Gaussian component. Provide the mathematical expression for $\boldsymbol{\mu}_k$ using the MLE.
4) (Score: 20 %) By using the MLE for the estimation of parameters of a Gaussian mixture model, do the resulting equations constitute a closed-form solution? If not, which scheme is used to solve the resulting set of equations and why?

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TOPIC 2

A classification problem is formed by two classes. We are given a set of 2-dimensional data $\mathbf{X} = [\mathbf{x}_1\ \mathbf{x}_2\ \mathbf{x}_3\ \mathbf{x}_4]$:

$$\mathbf{X} = \begin{bmatrix} 1 & 1 & 0 & -2 \\ 2 & 0 & -1 & -1 \end{bmatrix} \tag{1}$$

each belonging to one of the two classes, as indicated in the class label vector $\mathbf{l} = [l_1\ l_2\ l_3\ l_4]$:

$$\mathbf{l} = [1 \quad 1 \quad 2 \quad 2]. \tag{2}$$

Using the above (training) vectors and the corresponding class labels, classify the following vectors:

$$\mathbf{x}_5 = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \qquad \mathbf{x}_6 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad \mathbf{x}_7 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \qquad \mathbf{x}_8 = \begin{bmatrix} -2 \\ 0 \end{bmatrix} \tag{3}$$

using the following classifiers:

1) (Score: 25%) The Nearest Class Centroid (NCC) classifier.
2) (Score: 25%) The Nearest Neighbor Classifier using only one neighbor.

Moreover, use Fisher's Discriminant Analysis to:

1) (Score: 25%) Calculate the projection vector $\mathbf{w} \in \mathbb{R}^2$ obtained by the training data.
2) (Score: 25%) Draw in Figure 2 the projection vector of LDA $\mathbf{w}$ calculated in question 1.

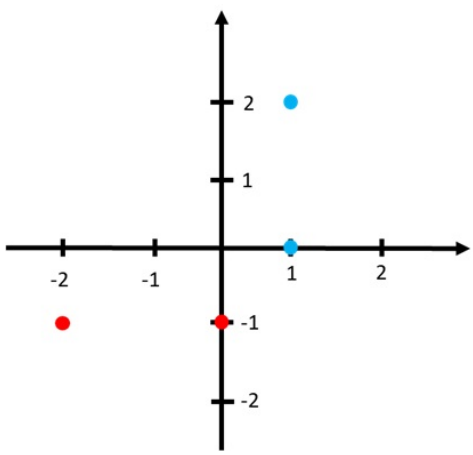

Fig. 2. 2-dimensional data points plotted as dots. Data belonging to class 1 are illustrated in blue color. Data belonging to class 2 are illustrated in red color.

TOPIC 3

Kernel functions $k(\mathbf{x}, \mathbf{x}')$ over two vectors $\mathbf{x}$ and $\mathbf{x}'$ can be used to solve non-linearly separable problems using linear techniques.

Answer the following questions:

1) (Score: 20 %) When is a function of the form $k(\mathbf{x}, \mathbf{x}')$ a valid kernel function?
2) (Score: 50 %) A polynomial kernel is given by

$$k_1(\mathbf{x}, \mathbf{x}') = (\mathbf{x}^\top \mathbf{x}' + c)^2, \quad c > 0 \tag{4}$$

   Verify that the polynomial kernel in Eq. 4 is a valid kernel for a two-dimensional input space, $\mathbf{x} = [x_1, x_2]$.
3) (Score: 30 %) Another kernel function is given by

$$k_2(\mathbf{x}, \mathbf{x}') = \exp\left(k_1(\mathbf{x}, \mathbf{x}')\right) + k_1(\mathbf{x}, \mathbf{x}') \tag{5}$$

   Does Eq. 5 constitute a valid kernel? Explain why / why not.

TOPIC 4

You are given a set of data points $\mathbf{x}_i \in \mathbb{R}^2$, $i = 1, \ldots, 53$ plotted in Figure 3.

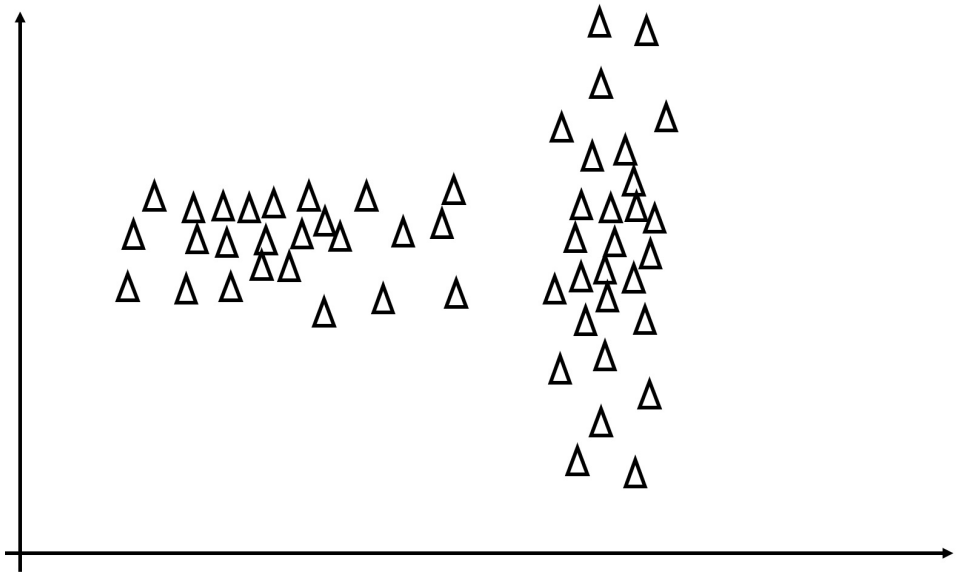

Fig. 3. 2-dimensional data points plotted as triangles.

Answer the following questions:

1) (Score: 30%) We would like to group the data points in $K = 2$ clusters by applying the K-Means method. The two stars are used as initial cluster means (at iteration 1) of the K-Means method. Draw the final positions of the cluster means (no accurate numerical values are needed) in Figure 4 and provide the cluster labels of the (test) vectors $\mathbf{y}$, $\mathbf{q}$ and $\mathbf{z}$.

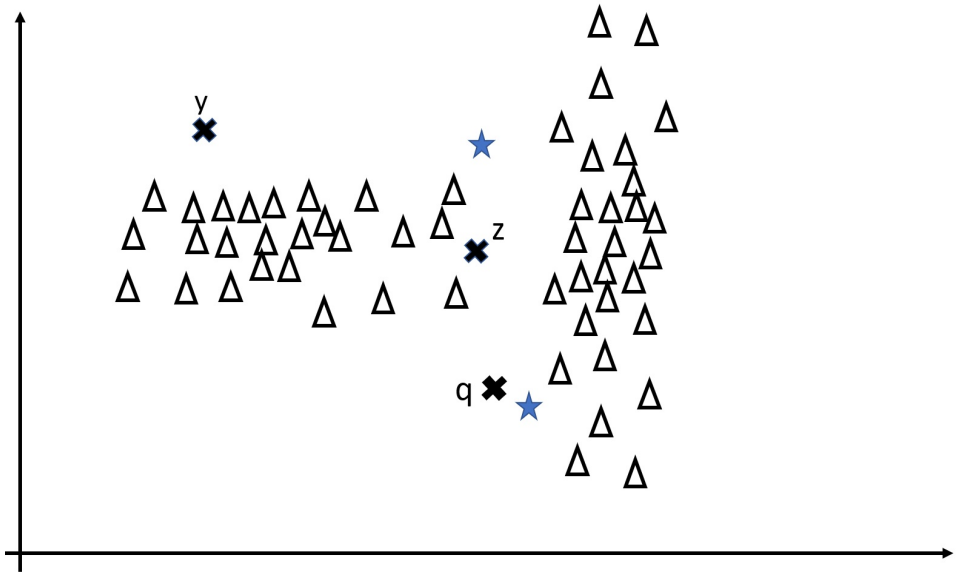

Fig. 4. 2-dimensional data points plotted as triangles. The initial cluster prototypes are illustrated as stars. Three test data points ($\mathbf{y}$, $\mathbf{q}$ and $\mathbf{z}$) are illustrated as crosses.

2) (Score: 20%) Explain why the initialization process of K-Means method plays a crucial role in the final clustering result.
3) (Score: 30%) We would like to group the data points in $K = 2$ clusters by applying the Mixture of Gaussians method. The two stars are used as initial means of the two Gaussian components of the mixture model (at iteration 1). Draw the final positions of the means of the two Gaussian components (no accurate numerical values are needed) in Figure 5 and provide the cluster labels of the (test) vectors $\mathbf{y}$, $\mathbf{q}$ and $\mathbf{z}$.
4) (Score: 20%) Explain which parameters of the Mixture of Gaussians method can (possibly) lead to a different clustering label assignment for the test data points $\mathbf{y}$, $\mathbf{q}$ and $\mathbf{z}$, compared to using K-Means (as in subquestion 2).

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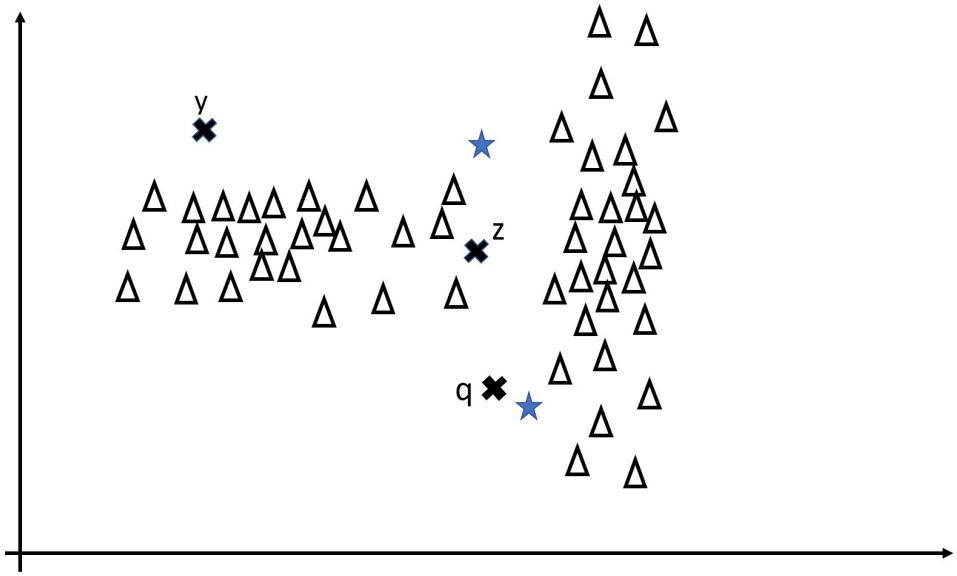

Fig. 5. 2-dimensional data points plotted as triangles. The initial cluster prototypes are illustrated as stars. Three test data points ($\mathbf{y}$, $\mathbf{q}$ and $\mathbf{z}$) are illustrated as crosses.